

Fretboard Scale Explorer

Teacher's Resource Pack

v2 — by Luke Blossom

Guitar · Bass · Ukulele · 14 Scales · Printable Worksheets · Practice Routines

What's Inside	
Scale Quick Reference	All 14 scales — step patterns, intervals, and genre guide
Fretboard Diagrams	6 guitar + 6 bass + ukulele scales shown in the key of A
New Features Guide	Dark mode, Position Box, Scale Comparison, Capo, Share & Print
Blank Worksheets	Printable grids for guitar, bass, and ukulele
Practice Routines	Week-by-week plans from beginner through intermediate
Tab Notation Guide	How to read and write guitar and bass tab
How to Use the Tool	Full classroom guide including all v2 features

Companion resource for the free Fretboard Scale Explorer web app.
Use alongside the interactive tool for best results.

What's New in Version 2

This updated Teacher's Resource Pack covers all features added since the original release. Every new feature below is available in the free web app at github.com/LukeBlossom/FretboardScaleExplorer.

■ Ukulele Support	A fourth instrument option — standard gCEA re-entrant tuning, Canadian aDf#B, and Low-G. All 14 scales work on ukulele. Scale diagrams are included in this pack.
■ Dark Mode	Press D on the keyboard or click the moon icon in the top-right to switch to dark mode. The preference is saved between sessions. Ideal for low-light practice rooms.
■ Position Box Filter	A new Position Box dropdown highlights a 5-fret window anchored to where the root note appears on the lowest string — the foundation of CAGED-system thinking. Notes outside the window fade out so students focus on one position at a time.
■ Scale Comparison Overlay	A second scale can be overlaid on the fretboard in purple. Notes in both scales are shown with a ring; notes only in the comparison scale appear as hollow purple dots. Ideal for showing the difference between, for example, Natural Minor and Dorian.
■ Capo Support	Set a capo from fret 1–7. The fretboard shifts accordingly — notes below the capo are greyed out and all pitch calculations adjust to the new effective open position.
■ Share Link	Click the link icon in the header to copy a URL that encodes the current key, scale, instrument, tuning, and capo. Paste it to share an exact setup with a student.
■■ Print Fretboard	Click the print icon (or use Ctrl+P / Cmd+P) to print just the fretboard — controls and backing track sections are hidden automatically for a clean printout.
■■ Keyboard Shortcuts	Shift+← / Shift+→ cycles the root note up or down. D toggles dark mode. Useful when presenting on a projector.

All features are free and work offline — no internet connection required after the page is loaded.

How to Use the Tool

The Fretboard Scale Explorer is a free, single-file web app. Open it in any modern browser — no installation or internet required. Works on laptop, tablet, or class projector.

1 — Choose your instrument

Select Guitar (6-string), Bass (4-string), Bass (5-string), or Ukulele (4-string). Tuning options and string layout update automatically.

2 — Set root note and scale

Choose any of the 12 chromatic root notes and one of 14 scales — from Pentatonic Minor to Lydian mode.

3 — Explore the fretboard

Red dots mark the root. Teal dots mark other scale tones. Click any dot to hear the note. Use Display to switch between Note Names, Intervals, and Scale Degrees.

4 — Change tuning

Guitar: Drop D, Open G, Open D, Half Step Down, Full Step Down. Bass: Drop D, Half Step Down, 5-string options. Ukulele: Standard gCEA, Canadian aDf#B, Low-G.

5 — Set a capo (new)

Choose Fret 1–7 from the Capo dropdown. Notes below the capo grey out and pitch calculations shift automatically.

6 — Focus on a position (new)

Use the Position Box dropdown to highlight a 5-fret CAGED-style window. Notes outside the window fade — helping students learn one position at a time.

7 — Compare two scales (new)

Choose a second scale from the Compare Scale dropdown. Purple dots show notes in the comparison scale. A ring on a teal dot means that note appears in both scales.

8 — Look up a song

Type any song into the Song Key Lookup box to set the fretboard to that song's key and scale. Over 270 songs are in the database.

9 — Find backing tracks

The YouTube Backing Track Generator builds a search query from your current key, scale, and tempo — with a built-in progression player and drum machine.

10 — Share or print (new)

Click the link icon to copy a shareable URL, or the print icon to print just the fretboard. Press D to toggle dark mode for low-light practice rooms.

Classroom Tips

- Display the tool on a projector — use Shift+← / Shift+→ to cycle root notes live.
- Use Position Box to isolate one CAGED shape — have students play only those frets.
- Use Compare Scale to show the difference between Natural Minor and Dorian in real time.
- Click notes to hear them. Have students predict the next scale tone before you click.
- Use Song Key Lookup to connect theory to songs students already know.
- Share a pre-configured URL with students so they open the exact key and scale for the lesson.

- Print the fretboard (print button) as a handout. Blank worksheets are at the back of this pack.
- For ukulele students: select Ukulele — all 14 scales apply in gCEA standard tuning.

Scale Quick Reference

All 14 scales in the tool. *Notes in C* shows each scale built from C. W = whole step, H = half step, m3 = minor third, A = augmented second.

Scale	Step Pattern	Notes in C	Common Genres
Major	W W H W W W H	C D E F G A B	Pop, Country, Classical
Natural Minor	W H W W H W W	C D D# F G G# A#	Rock, Metal, Classical
Harmonic Minor	W H W W H A H	C D D# F G G# B	Classical, Metal, Flamenco
Melodic Minor	W H W W W W H	C D D# F G A B	Jazz, Classical
Pentatonic Major	W W m3 W m3	C D E G A	Country, Blues, Pop
Pentatonic Minor	m3 W W m3 W	C D# F G A#	Rock, Blues, Metal
Blues	m3 W H H m3 W	C D# F F# G A#	Blues, Rock, R&B
Dorian	W H W W W H W	C D D# F G A A#	Jazz, Funk, Celtic
Phrygian	H W W W H W W	C C# D# F G G# A#	Metal, Flamenco
Lydian	W W W H W W H	C D E F# G A B	Film, Jazz, Prog Rock
Mixolydian	W W H W W H W	C D E F G A A#	Rock, Blues, Country
Locrian	H W W H W W W	C C# D# F F# G# A#	Metal, Avant-garde
Whole Tone	W W W W W W	C D E F# G# A#	Jazz, Impressionist
Diminished	W H W H W H W H	C D D# F F# G# A B	Jazz, Classical, Metal

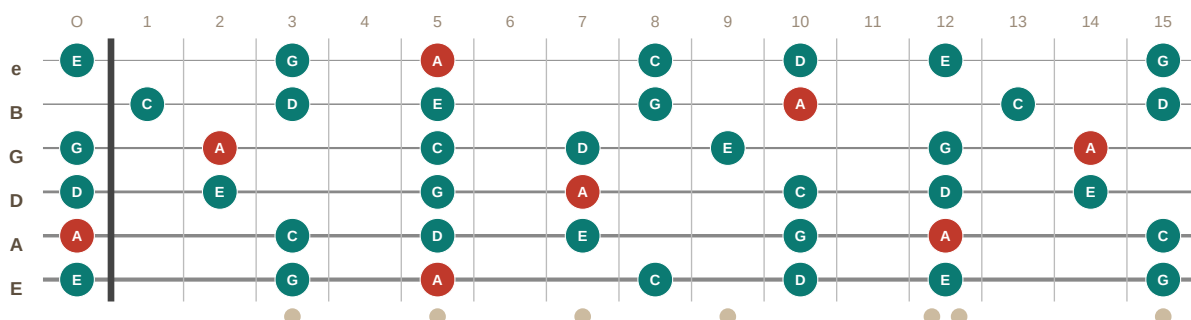
Interval Name Reference

Symbol	Name & Meaning
R	Root — the tonal centre of the scale
b2 / 2	Minor / Major 2nd — 1 or 2 semitones above root
b3 / 3	Minor / Major 3rd — the key factor in "minor" vs "major" sound
4 / b5	Perfect Fourth / Tritone (the "blue note")
5 / b6	Perfect Fifth (strong, stable) / Minor Sixth (dark)
6 / b7	Major Sixth / Minor Seventh — common in blues and dominant chords
7	Major Seventh — bright, jazzy; one half step below the octave

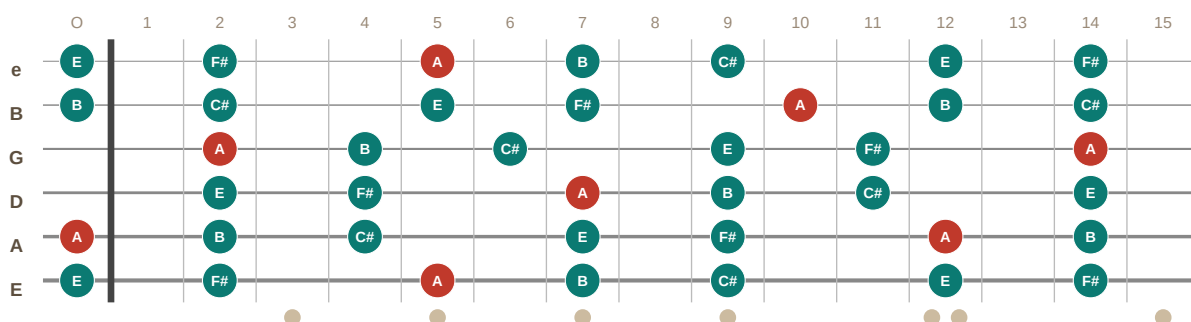
Guitar Scale Diagrams — Key of A

Six essential scales shown on a standard-tuned guitar (EADGB_e), frets 1–12. **Red dots** mark the root note (A). **Teal dots** mark other scale tones.

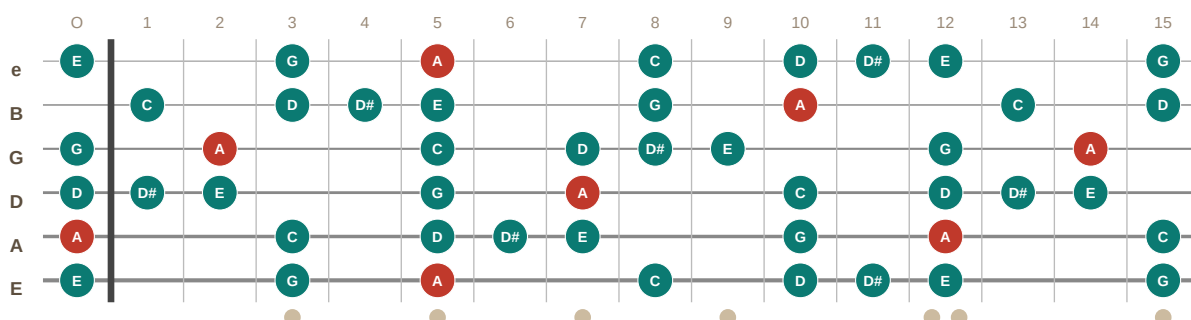
A Pentatonic Minor — The most-used scale in rock, blues, and metal



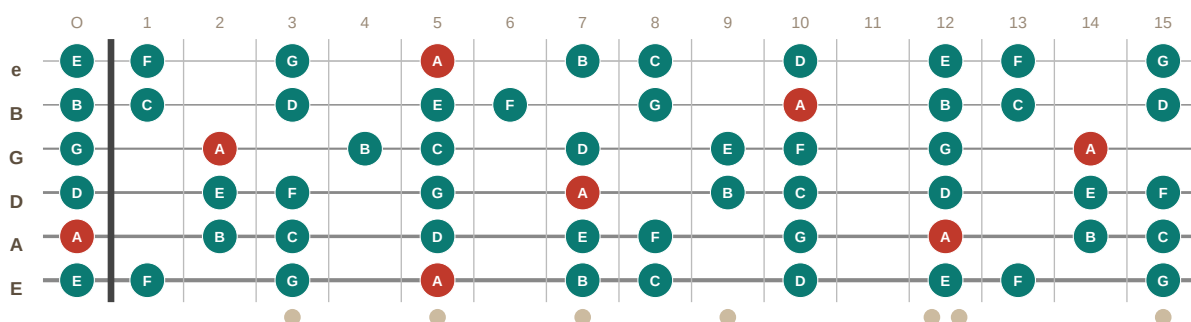
A Pentatonic Major — Bright, country and pop flavour



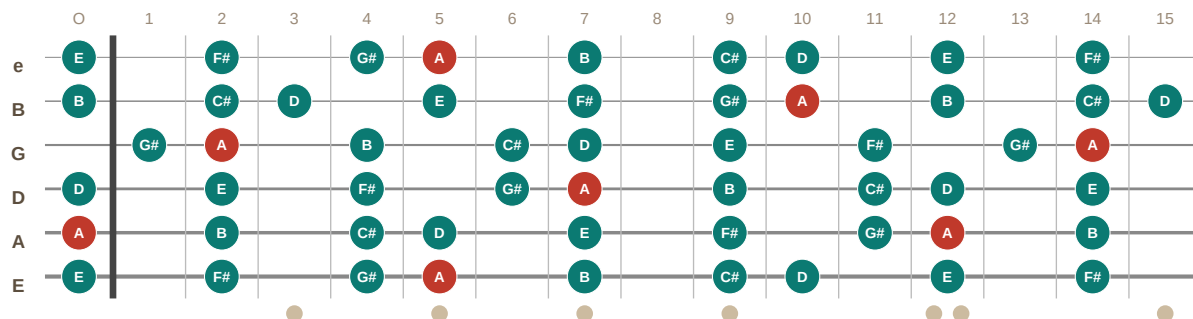
A Blues — Pentatonic minor + the flat 5 "blue note"



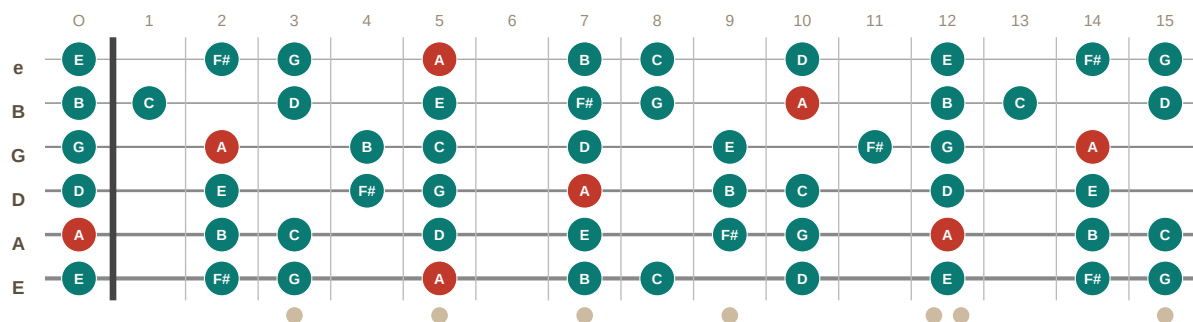
A Natural Minor — Foundation of minor key playing (Aeolian mode)



A Major — Bright, foundational scale for pop and country



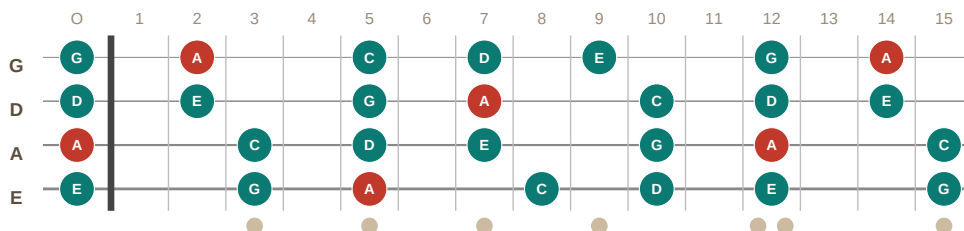
A Dorian — Natural minor with a raised 6th; jazz, funk, and Celtic



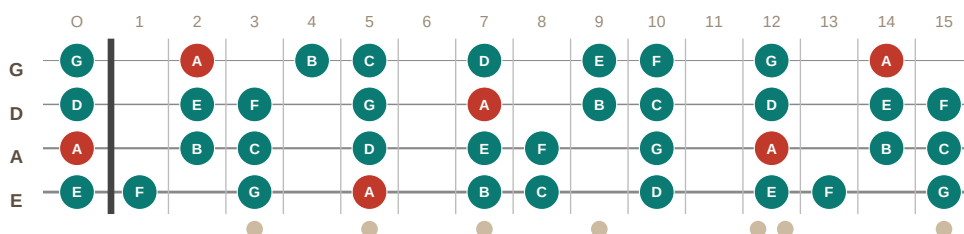
Bass Scale Diagrams — Key of A

Six key scales for 4-string bass in standard tuning (EADG), frets 1–12. The same scale patterns apply as guitar — identical intervals, lower range.

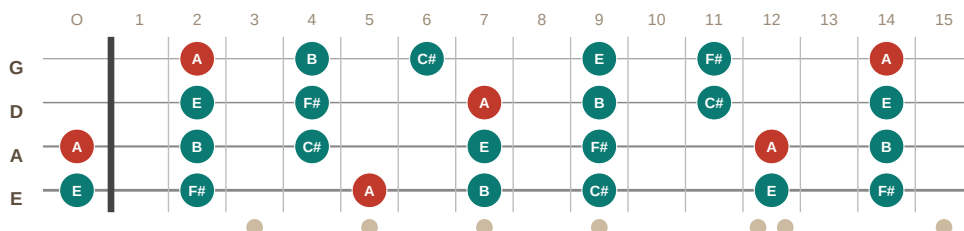
A Pentatonic Minor — Rock, funk, and R&B; bass lines



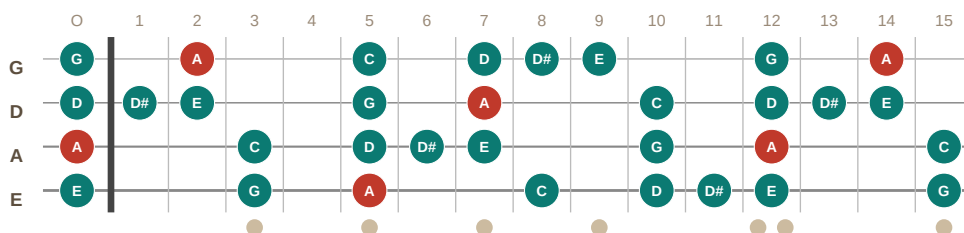
A Natural Minor — Foundation of minor key bass playing



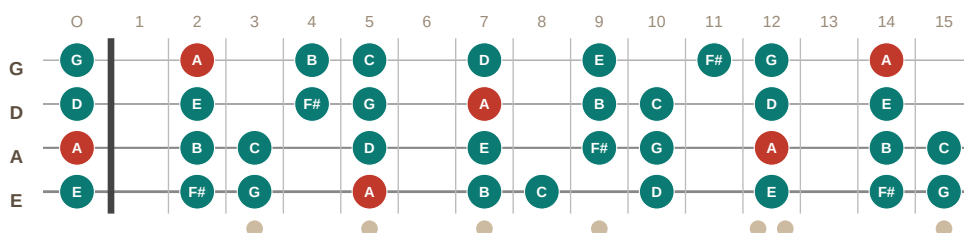
A Pentatonic Major — Country, pop, and gospel bass lines



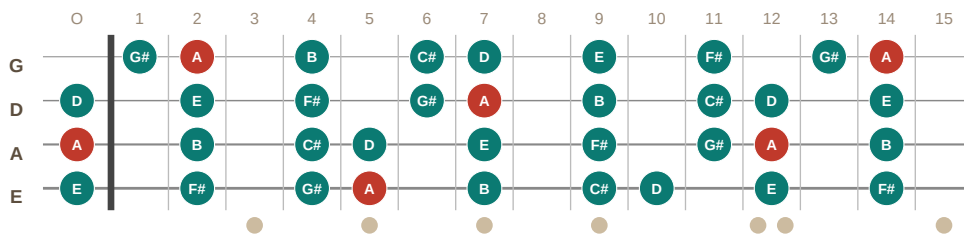
A Blues — Walking blues bass and improvisation



A Dorian — Funk and jazz fusion; iconic for bass solos



A Major — Pop, country, and major key walking bass



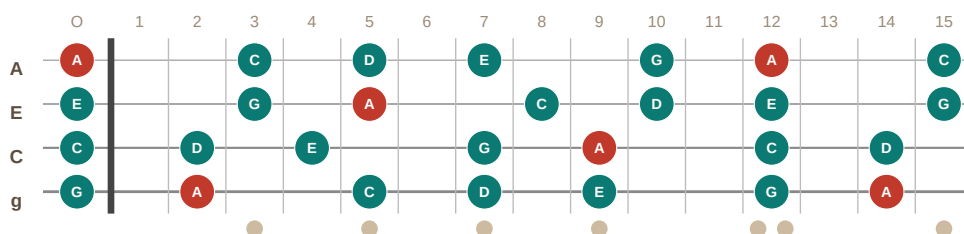
Ukulele Scale Diagrams — Key of A

Four scales for ukulele in standard gCEA re-entrant tuning, frets 1–12. The g string (top) is tuned to G4 — *higher* than the C string below it. This re-entrant tuning is what gives ukulele its characteristic bright, jangly sound. The O column shows open string notes.

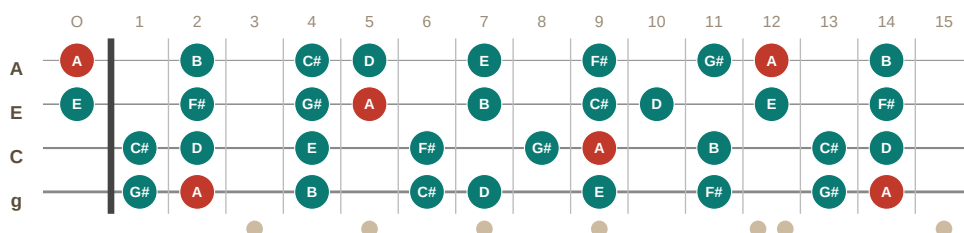
Standard gCEA tuning (low → high by pitch): C4 · E4 · G4 · A4

Displayed top to bottom on the diagram: g (G4) · C (C4) · E (E4) · A (A4)

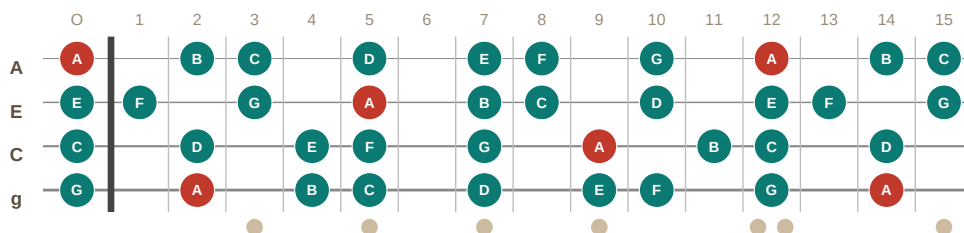
A Pentatonic Minor — The go-to scale for ukulele improvisation in any style



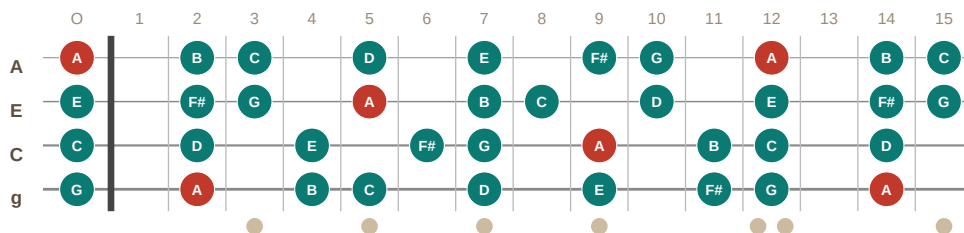
A Major — Bright and cheerful; great for Hawaiian and pop ukulele



A Natural Minor — Melancholic character; suits ballads and folk



A Dorian — Minor with brightness; jazz, Celtic, and contemporary ukulele



Selecting Ukulele in the Tool

- Choose  Ukulele (4-string) from the Instrument dropdown.
- Three tuning options: Standard gCEA, Canadian aDf#B (up a whole step), and Low-G (replaces re-entrant g with a lower G3 string).

- All 14 scales, the Position Box filter, and the Compare Scale overlay all work with ukulele exactly as they do for guitar and bass.
- The audio playback correctly reflects the re-entrant g string pitch.

Position Box Filter & Scale Comparison

The Position Box Filter

The **Position Box** dropdown (second control row, below the main controls) highlights a 5-fret window on the fretboard — one "box" position of the scale. Notes outside the window fade out so students can focus on one shape at a time.

How positions are calculated: The tool finds every fret on the lowest string where the root note appears, then creates a 5-fret window starting from that fret. For A Pentatonic Minor in standard tuning, the low E string plays an A at fret 5 and fret 17 — so two positions appear in the dropdown.

Classroom exercise — CAGED positions

- Set the fretboard to A Pentatonic Minor. Select **Pos 1 (frets 5–9)**.
- Have students play only the highlighted notes in order, low string to high.
- Switch to Pos 2 (frets 17–21) — same notes, different area of the neck.
- Challenge: play from one position into the next without stopping.
- Print the blank worksheets (page 12) and have students draw the position shapes.

Scale Comparison Overlay

The **Compare Scale** dropdown (second control row) overlays a second scale on the fretboard using purple dots. This lets you show how two scales relate — which notes they share, and which are different.

Reading the overlay:

- **Teal dot** — note in the main scale only
- **Purple dot** — note in the compare scale only
- **Teal dot with purple ring** — note in both scales

Suggested comparisons for the classroom

Main Scale	Compare Scale	What it shows
Pentatonic Minor	Natural Minor	The 2 extra notes in natural minor (b2 and b6) appear in purple
Natural Minor	Dorian	Dorian's raised 6th lights up purple — one note difference
Major	Mixolydian	The flat 7th in Mixolydian (the "bluesy" note) appears in purple
Pentatonic Minor	Blues	The blue note (b5) appears as the one extra purple dot
Natural Minor	Harmonic Minor	The raised 7th that creates tension before the root appears in purple

Blank Fretboard Worksheets

Print this page. Students use these grids to map out scale positions, chord tones, or their own note discoveries.

Guitar — Standard Tuning (E A D G B e) — Frets 1–12

Student: _____ Scale: _____ Key: _____ Date: _____

	1	2	3	4	5	6	7	8	9	10	11	12
e												
B												
G												
D												
A												
E												

Student: _____ Scale: _____ Key: _____ Date: _____

	1	2	3	4	5	6	7	8	9	10	11	12
e												
B												
G												
D												
A												
E												

Bass — Standard Tuning (E A D G) — Frets 1–12

Student: _____ Scale: _____ Key: _____ Date: _____

	1	2	3	4	5	6	7	8	9	10	11	12
G												
D												
A												
E												

Student: _____ Scale: _____ Key: _____ Date: _____

	1	2	3	4	5	6	7	8	9	10	11	12
G												
D												
A												
E												

Ukulele Worksheets

Standard gCEA re-entrant tuning. The g string is at the top of the diagram.

Ukulele — Standard Tuning (g C E A) — Frets 1–12

Student: _____ Scale: _____ Key: _____ Date: _____

	1	2	3	4	5	6	7	8	9	10	11	12
A												
E												
C												
g												

Student: _____ Scale: _____ Key: _____ Date: _____

	1	2	3	4	5	6	7	8	9	10	11	12
A												
E												
C												
g												

Student: _____ Scale: _____ Key: _____ Date: _____

	1	2	3	4	5	6	7	8	9	10	11	12
A												
E												
C												
g												

Practice Routines

Designed to be used alongside the Fretboard Scale Explorer. Each level builds on the previous. Suggested session length: 20–30 minutes.

Beginner — Building the Foundation (Weeks 1–6)

Goal: Know A Pentatonic Minor in one position. Play confidently over a slow backing track.

Weeks 1–2: Learn the Shape

- Open the tool, select A, Pentatonic Minor.
- Open the Position Box dropdown and choose **Pos 1 (frets 5–9)**. This highlights the classic "box 1" shape — the most common starting position.
- Practice ascending and descending slowly with a metronome at 60 BPM.
- Click each teal dot in the tool to hear the note before playing it.

Weeks 3–4: Add Rhythm

- Use the Backing Track Generator: set A, Pentatonic Minor, 70–80 BPM, then search YouTube.
- Play through the box 1 shape along with the track — not improvising yet.
- Always end phrases on the root note (red dot). Notice how it sounds resolved.

Weeks 5–6: Start Improvising

- Stay in position 1 but choose 3–4 notes at random to create short phrases.
- Switch Display to Intervals — identify which note feels "at home" (R = root).
- Try switching to Position 2 in the Position Box to explore a new area of the neck.

Intermediate — Expanding the Palette (Weeks 7–16)

Goal: Know all pentatonic positions, understand modes, and use the comparison tool to see how scales relate.

Weeks 7–8: The Five Pentatonic Positions

- Set A Pentatonic Minor. Cycle through all positions in the Position Box dropdown.
- Learn each position shape. Print blank worksheets and draw the dots by hand.
- Practice connecting adjacent positions in one smooth run.

Weeks 9–10: Natural Minor vs. Pentatonic Minor

- Set main scale to Natural Minor. In Compare Scale, choose Pentatonic Minor.
- Purple dots show the two extra notes (b2 and b6) that the full minor scale adds.
- Improvise mostly in pentatonic, then target a purple note for colour.

Weeks 11–12: Modes with the Comparison Overlay

- Set Natural Minor, Compare: Dorian. One note lights up purple — the raised 6th. That single note gives Dorian its characteristic sophistication.
- Set Major, Compare: Mixolydian. The flat 7th "bluesy" note appears in purple.
- Use Song Key Lookup: "So What" (Miles Davis) for Dorian; "Sweet Home Alabama" for Mixolydian.

Weeks 13–16: Key Changes and Song Application

- Pick three songs from Song Key Lookup in different keys (E, A, G).
- Use the Share Link button to send students a pre-configured fretboard URL.

- Challenge: find the same scale shape in three different positions for each key.
- For ukulele students: switch to Ukulele mode and repeat weeks 1–8 in gCEA tuning.

Introduction to Reading Tab Notation

Guitar tablature (tab) is a simple way to write music for fretted instruments without reading traditional notation. Numbers tell you which fret to press on which string.

How the Tab Staff Works

A guitar tab staff has six horizontal lines — one per string. The bottom line is the thickest (low E), the top line is the thinnest (high e). Bass tab uses four lines (E, A, D, G). Ukulele tab uses four lines (g, C, E, A).

Reading the Numbers

0 = open string (no fret pressed). Any other number = press that fret. Numbers stacked vertically = play simultaneously (chord). Numbers in sequence = play one at a time (melody or riff).

Common Tab Symbols

Symbol	Meaning	How to Play
0	Open string	Pick the string without pressing any fret
1–24	Fret number	Press at that fret and pick the string
x	Muted note	Touch the string lightly — a click, not a pitch
h	Hammer-on	5h7 — pick fret 5, then hammer finger onto fret 7
p	Pull-off	7p5 — pick fret 7, pull finger off to sound fret 5
b	Bend	7b9 — pick fret 7, push string sideways to raise pitch
/	Slide up	5/7 — slide from fret 5 to 7 while holding pressure
\	Slide down	7\5 — same, but slide toward the nut
~	Vibrato	Bend and release rapidly for an expressive wobble

Quick Tips for Students

- Always use the Fretboard Scale Explorer to hear what a scale sounds like before trying to read or write tab for it.
- Tab does not show rhythm by default — listen to a recording or backing track to hear where the beats fall.
- Use blank fretboard worksheets to map positions, then write tab using the notes you find.
- Start with single-string melodies before attempting multi-string runs.

Quick Reference — Scale Moods and Uses

A cut-out reference card. Shows the emotional character of each scale and the genres where it appears most.

Scale	Mood / Character	Who Uses It
Pentatonic Minor	Gritty, soulful, versatile	Rock, Blues, Metal, R&B
Pentatonic Major	Bright, uplifting, simple	Country, Pop, Gospel
Blues	Tense, emotional, expressive	Blues, Rock, Jazz
Major	Happy, resolved, foundational	Pop, Classical, Country
Natural Minor	Sad, dark, dramatic	Rock, Classical, Film
Harmonic Minor	Exotic, tense, operatic	Classical, Metal, Flamenco
Dorian	Minor with brightness, sophisticated	Jazz, Funk, Celtic, Fusion
Phrygian	Dark, Spanish, menacing	Metal, Flamenco, Spanish
Mixolydian	Major with a bluesy edge	Rock, Blues, Country, Jazz
Lydian	Dreamy, floating, cinematic	Film, Prog Rock, Jazz
Locrian	Tense, unstable, rarely resolved	Metal, Avant-garde
Whole Tone	Ambiguous, dreamlike, no strong pull	Jazz, Impressionist
Diminished	Tense, symmetrical, unsettling	Jazz, Classical, Metal

Fretboard Scale Explorer is a free resource by Luke Blossom. Share it freely with students and colleagues.
github.com/LukeBlossom/FretboardScaleExplorer · ko-fi.com/lukeblossom